INCH-POUND

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DETAIL SPECIFICATION

EXPLOSIVE SEQUENCE CONTROL, DCU-241/A

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the explosive sequence control (ESC), DCU-241/A.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-D-21625 - Design and Evaluation of Cartridges for Cartridge Actuated Devices

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS

30003-850AS135 - Explosive Sequence Control, DCU-241/A

30003-850AS804 - Test Fixture, Explosive Sequence Control, DCU-241/A

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1377

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30003-850AS852 - Vibration Fixture, Explosive Sequence Control, DCU-241/A

30003-850AS853 - Spacer

(Copies of these documents are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch [Code E26], 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquires, or problems to nswc_iheodtd_dmo@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM E1742/E1742M - Standard Practice for Radiographic Examination

(Copies of this document are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AS5604 - Compass; Magnetic, Pilot's Standby

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Design and construction. The ESC and all of its components shall be manufactured in accordance with NAVAIR Drawing 30003-850AS135. The selector shall be designed to minimize weight and shall not exceed 1.25 pounds.

3.2.1 Torque requirements of threaded joints. The torque requirements of subassemblies of the device that contains the provisions for the attachment of shielded mild detonating cord (SMDC) or for the mounting or installation of the device shall be able to withstand a minimum of 400 inch-pounds of breakaway torque for thread diameters of 1 inch or larger, and a minimum value of one-half the maximum torque value specified in table I for thread sizes less than 1 inch in diameter. The torque requirement of other subassemblies of the device which could be loosened or disassembled by handling, vibration, shipment, etc., shall withstand a minimum value of 200 inch-pounds for thread diameters of 1 inch or larger and shall withstand a minimum value of one-third of the maximum torque value specified in table I for thread sizes under 1 inch in diameter.

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TABLE I. Breakaway torque.

Bolt or Screw Size	Maximum Torque (inch-pounds)
No. 4-40 UNC-3A	3
No. 6-32 UNC-3A	6
No. 8-32 UNC-3A	9
No. 10-32 UNF-3A	13
1/4-28 UNF-3A	30
5/16-24 UNF-3A	60
3/8-24 UNF-3A	80
7/16-20 UNF-3A	100
1/2-20 UNF-3A	150
9/16-18 UNF-3A	200
5/8-18 UNF-3A	300
3/4-16 UNF-3A	400
7/8-14 UNF-3A	600

3.3 Materials. All materials used in the manufacture of the ESC shall conform to the standards established at technical service release as specified (see 6.2). Any divergence shall require technical evaluation and potential design certification testing.

3.4 ESC shaft. The ESC shaft shall consist of an HNS type I transfer booster charge, in accordance with NAVAIR Drawing 30003-850AS135, which aligns with the appropriate SMDC outlet port to propagate explosive energy to the selected output SMDC port without detonation of the other output SMDC ports.

3.4.1 Visual inspection. All ESC shafts manufactured shall be capable of passing a visual inspection and a screening for defects as outlined in 4.6.

3.4.2 Leakage. ESC shafts shall not exhibit a leak rate in excess of 1×10^{-5} cubic centimeters per second of air when tested in accordance with 4.7.

3.4.3 Radiographic requirement. ESC shafts shall be free of imperfections when examined radiographically in accordance with 4.8.

3.4.4 Shaft gap. The explosive loaded shaft subassembly shall be capable of detonating an SMDC test tip across a gap which is four times the maximum production gap when tested in accordance with 4.9.

3.5 ESC.

3.5.1 Visual inspection. Each ESC manufactured shall be capable of passing a visual inspection and a screening for defects as outlined in 4.6.

3.5.2 Radiographic requirement. ESCs shall be free of imperfections in assembly when examined radiographically in accordance with 4.8.

3.5.3 Demagnetization. ESCs shall not deflect an approved pilot's standby compass more than three degrees in either direction when tested in accordance with 4.10.

3.5.4 Actuation force. The total linear force required to position the handle from normal to aft initiates shall be 20 to 30 pounds. The torque required to release the handle from the aft initiates position shall be 10 to 20 inch-pounds. See 4.11 for more information.

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3.6 Environmental requirements

3.6.1 Lifecycle. The ESC shall not fire or incur damage or degradation when subjected to 3,000 cycles of the selector handle when tested in accordance with 4.12.

3.6.2 Temperature/shock/humidity/altitude (TSHA). The ESC shall not fire or incur damage or degradation when subjected to TSHA testing in accordance with MIL-D-21625 and 4.13.

3.6.3 Vibration. The ESC shall not fire or incur damage or degradation when subjected to vibration testing in accordance with 4.14.

3.6.4 Shock. The ESC shall not fire or incur damage or degradation when subjected to shock testing in accordance with MIL-D-21625 and 4.15.

3.7 Ballistic requirements. When tested as specified in 4.16, the ESC shall initiate the specified outlet SMDC test tip based on the operating mode as defined by the handle position, as shown in table II. The remaining two SMDC test tips shall not initiate, as shown in table II.

TABLE II. SMDC test tip results.

Handle Position	Inlet SMDC Test Tip Initiated (Port No.)	Outlet SMDC Test Tip Propagated (Port No.)	Outlet SMDC Test Tip Not Propagated (Port No.)
Normal	1	2	3 and 4
Solo	1	4	2 and 3
Aft	1	3	2 and 4

3.7.1 High order detonation. The SMDC output shall achieve high order detonation. The minimum plate dent for high order detonation of the output SMDC test tip on the witness plate shall be 0.001 inch.

3.7.2 Misfire. There shall be no misfires.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Production inspection (see 4.3).

c. Lot acceptance inspection (see 4.4).

4.2 First article inspection. Before entering into quantity production, an acceptable first article sample of 3 shaft subassemblies and 24 ESCs shall be prepared using the same procedures and methods proposed for the production lot. The first article sample shall be subjected to the tests listed in table III. The contractor shall test the 3 shaft subassemblies and deliver the ESCs to the activity designated by the contracting agency for sequential tests. Environmental exposure shall be conducted sequentially as shown in table III. First article testing is for the purpose of determining that the contractor, prior to starting production, is capable to producing items that comply with the technical requirements as specified. As specified (see 6.2), a first article inspection is required if two or more years have passed since the item was manufactured, after major turnover in personnel, following major changes to the production facility, or following transfer of production to a new facility or site. Failure of any shaft subassembly or ESC to meet the requirements of table III shall be cause for rejection of the first article represented.

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TABLE III. First article inspection and environmental test sequence.

Test Sequence	Requirement	Requirement Paragraph	Test Paragraph	Test Group Quantity			
				9 ESC	9 ESC	6 ESC	3 Shaft
1	Breakaway torque	3.2.1	4.5	9	9	6	3
2	Visual inspection	3.4.1, 3.5.1	4.6	9	9	6	3
3	Leakage	3.4.2	4.7	---	---	---	3
4	Radiographic examination	3.4.3, 3.5.2	4.8	9	9	6	3
5	Gap test	3.4.4	4.9	---	---	---	3
6	Demagnetization	3.5.3	4.10	9	9	6	---
7	Actuation force	3.5.4	4.11	9	9	6	---
8	Lifecycle	3.6.1	4.12	9	---	---	---
9	TSHA	3.6.2	4.13	---	---	0	---
	Handle position normal (in)			4	4	---	---
	Handle position aft (out)			5	5	---	---
10	Visual inspection	3.5.1	4.6	9	9	---	---
11	Non-gunfire vibration	3.6.3	4.14.1	---	---	---	---
	Handle position normal (in)			4	4	---	---
	Handle position aft (out)			5	5	---	---
12	Gunfire vibration	3.6.3	4.14.2	---	---	---	---
	Handle position normal (in)			4	4	---	---
	Handle position aft (out)			5	5	---	---
13	Shock	3.6.4	4.15	---	---	---	---
	Handle position normal (in)			4	4	---	---
	Handle position aft (out)			5	5	---	---
14	X-ray Examination	3.5.2	4.8.2.1	---	---	---	---
	Handle position normal (in)			4	4	---	---
	Handle position aft (out)			5	5	---	---
	Functional Testing						
15	High temperature (+200 °F)	3.7	4.16	---	---	---	---
	Handle position normal (in)			1	1	1	---
	Handle position aft (out)			1	1	1	---
	Handle position solo (spacer)			1	1	---	---
15	Low temperature (-65 °F)	3.7	4.16	---	---	---	---
	Handle position normal (in)			1	1	---	---
	Handle position aft (out)			1	1	1	---
	Handle position solo (spacer)			1	1	1	---
15	Ambient temperature (+65° to +95 °F)	3.7	4.16	---	---	---	---
	Handle position normal (in)			1	1	1	---
	Handle position aft (out)			1	1	---	---
	Handle position solo (spacer)			1	1	1	---

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4.3 Production inspection. All production shafts and ESCs shall be inspected and screened to the examinations and tests listed in table IV.

TABLE IV. Production inspection.

Item	Test	Requirement Paragraph	Test Paragraph
Shaft	Visual inspection	3.4.1	4.6
Shaft	Leakage	3.4.2	4.7
Shaft	Radiographic examination	3.4.3	4.8
ESC	Visual inspection	3.5.1	4.6
ESC	Radiographic examination	3.5.2	4.8
ESC	Demagnetization	3.5.3	4.10
ESC	Actuation force	3.5.4	4.11

4.4 Lot acceptance inspection. Lot acceptance shall consist of the examinations and tests specified in table V. Failure of any sample to comply with the requirements of table V shall be cause for rejection of the lot represented.

TABLE V. Lot acceptance inspection.

Inspection/Test	Requirement Paragraph	Test Paragraph	Sample
Breakaway torque	3.2	4.5	All
Visual inspection	3.5.1	4.6	All
Radiographic examination	3.5.2	4.8	All
Demagnetization	3.5.3	4.10	All
Actuation force	3.5.4	4.11	All
Functional (-65 °F, +70 °F, +200 °F)	3.7	4.16	
Handle position normal (in)			⅓
Handle position aft (out)			⅓
Handle position solo (spacer)			⅓

4.4.1 Sampling. Samples for ballistic test shall be selected from the production lot in accordance with the requirements of general inspection level II of ASQ/ANSI Z1.4. Three additional ESCs shall be provided with each test sample for investigative purposes.

4.5 Breakaway torque. Each first article test sample and lot acceptance test sample shall be inspected by performing breakaway torque testing to determine that the joint meets the requirements specified in 3.2.1.

4.6 ESC and ESC shaft visual inspection. All ESC shafts shall be visually examined prior to ESC assembly and all ESCs shall be visually examined after assembly. Those ESCs with inadequate or illegible descriptive markings or serial numbers, dents, deep scratches, or other defects which may prevent proper installation or function shall be rejected.

4.7 ESC shaft leak test. Each ESC shaft shall be leak tested in a dry gas leak tester to determine conformance to 3.4.2. Units which exhibit a leak rate in excess of $1x10^{-5}$ standard cubic centimeters per second of air at a pressure differential of 1.0±0.1 atmosphere shall be rejected.

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4.8 Radiographic examination.

4.8.1 ESC shaft examination. All ESC shafts shall be identified with serial numbers prior to radiographic examination. The ESC shafts shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the ESC shafts displayed thereon. The ESC shafts shall be positioned to give the most revealing cross section of the shaft. The radiograph identification shall include the complete lot number for the ESC shafts and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the radiographic plate. Defective shafts shall be clearly identified on the radiographic plate and those defective shall be removed from the production lot. Radiographs or copies of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test as specified (see 6.2).

4.8.1.1 X-ray examination. Each ESC shaft shall be X-rayed prior to ESC assembly, in accordance with ASTM E1742/E1742M, to confirm the presence and proper assembly of all parts. Any observable imperfections in assembly shall be cause for rejection.

4.8.1.2 N-ray examination. Each ESC shaft shall be N-rayed, prior to ESC assembly, to confirm the presence of explosive, non-metal parts, no foreign hydrogenous materials, and the proper assembly of pyrotechnic/non-metallic materials. Any observable imperfection shall be cause for rejection.

4.8.2 ESC examination. All assembled ESCs shall be identified with serial numbers prior to radiographic examination. The ESCs shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the ESCs displayed thereon. The ESCs shall be positioned to give the most revealing cross-section of the ESC. The radiograph identification shall include the complete lot number, as stamped on the ESCs, and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the radiographic plate. Defective ESCs found by the vendor shall be clearly identified on the radiographic plate, and those found defective shall be removed from the production lot. Radiographs or copies of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test as specified (see 6.2).

4.8.2.1 X-ray examination. Each assembled ESC shall be X-rayed in accordance with ASTM E1742/E1742M to confirm the presence and proper assembly of all parts. Any observable imperfections in assembly shall be cause for rejection.

4.9 Shaft gap. Prior to assembly of the selector, shafts shall be selected at random, as defined in 4.2, and subjected to ambient temperature testing at a distance four times the maximum production gap.

4.10 Demagnetization. Each ESC shall be moved past a pilot's standby compass face (in accordance with SAE AS5604) by placing the ESC 6 inches from the compass face in a north-south horizontal position with the compass set in an east-west heading. The ESC and the compass shall be placed at the same height from the floor. The ESC shall be moved past the compass face in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved back past the compass face. This procedure shall be repeated for each 90-degree rotation of the ESC.

4.11 Actuation force. Each sequencer shall be cycled from the normal position to the aft initiates position, locked in place, and then returned to the normal position. The actuation force and torque shall be in accordance with 3.5.4.

4.12 Lifecycle. First article test specimens specified in table III shall be exposed to physical lifecycle testing of the selector. The selector handle shall be cycled 3,000 times, each cycling representing the full drawn-out position of the handle, then a 90-degree turn of the handle, the rotation back 90 degrees, and the release of the handle to its original position. The items shall not fire or incur damage or degradation during or subsequent to the lifecycle test.

4.13 Temperature/shock/humidity/altitude (TSHA) cycling. First article test specimens specified in table III shall be exposed to a 4-week (28-day) TSHA cycling environment as described below, in accordance with MIL-D-21625.

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The test items with stainless steel plugs and seals installed in the inlet and outlet ports shall be submitted to the TSHA cycling test outlined in table VI and the following requirements:

a. The test items shall be positioned in the test chamber vertically with the handle upward.
b. The handle position for the testing shall be as specified in table III.
c. No test item shall be in contact with another test item during the cycling shown in table VI.
d. There shall be no interim-withdraws of units.

TABLE VI. TSHA schedule.

Monday	0800	Place sequencers in chamber at 70 °F at 50 percent relative humidity (RH).
	1200	Raise chamber temperature to 160 °F and at 95 percent RH. The chamber shall reach 160 °F no later than 1300.
	1600	Remove the sequencers from above chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 70,000 feet (0.65 pounds per square inch [psi]).
Tuesday	0800	Remove the sequencers from above chamber and immediately place in chamber maintained at 70 °F at 50 percent RH.
	1200	Raise chamber temperature to 160 °F and at 95 percent RH. The chamber shall reach 160 °F no later than 1300.
	1600	Remove sequencers from above chamber and immediately place in a chamber maintained at 160 °F at 95 percent RH.
Wednesday	0800	Reduce chamber temperature to 70 °F at 50 percent RH. The chamber temperature shall reach 70 °F at 50 percent RH no later than 0900.
	1200	Raise chamber temperature to 160 °F and at 95 percent RH. The chamber temperature shall reach 160 °F no later than 1300.
	1600	Remove the sequencers from above chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 70,000 feet (0.65 psi).
Thursday	0800	Remove the sequencers from above chamber and immediately place in a chamber maintained at 70 °F at 50 percent RH.
	1200	Remove the sequencers from above chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 70,000 feet (0.65 psi).
	1600	Remove the sequencers from above chamber and immediately place in a chamber maintained at 160 °F at 95 percent RH.
Friday	0800	Reduce chamber temperature to 70 °F and at 50 percent RH. The chamber temperature shall reach 70 °F at 50 percent RH no later than 0900.
	1200	Raise chamber temperature to 160 °F and at 95 percent RH. The chamber temperature shall reach 160 °F no later than 1300.
	1600	Remove the sequencers from above chamber and immediately place in a chamber maintained at -65 °F at a standard ambient pressure.
NOTE: This schedule shall be followed for a total of 4 weeks (28 days), except that on the second and fourth weekend, the soak time shall be from 1200 on Friday until 0800 Monday, at a temperature of 160 °F at 95 percent.		

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4.13.1 Success criteria. The items shall not incur detrimental degradation or damage during the TSHA test and shall not allow moisture intrusion. Evidence of corrosion on internal components is not acceptable. When TSHA testing is complete, items shall be subjected to visual inspection prior to vibration testing. Port plugs shall be removed for this inspection.

4.14 Vibration test. Vibration test shall be performed using a fixture in accordance with NAVAIR Drawing 30003-850AS852.

4.14.1 Non-gunfire random and sinusoidal requirements. The combined random and sinusoidal (non-gunfire) requirements involve a combination of sinusoidal cycling (5 to 50 hertz) and high frequency (100 to 2,000 hertz) random vibration. The procedures shall be conducted along all three orthogonal axes at the following temperatures: -65 °F, +70 °F, and +200 °F. The non-gunfire random and sinusoidal vibration tests shall be performed on each of the ESCs. Two accelerometers shall be used to monitor the response of the ESCs during the test. One accelerometer shall be located on a handle and the other on a body of one of the ESCs, along the centerline of port number 4. The handle position for the testing shall be as specified in table III.

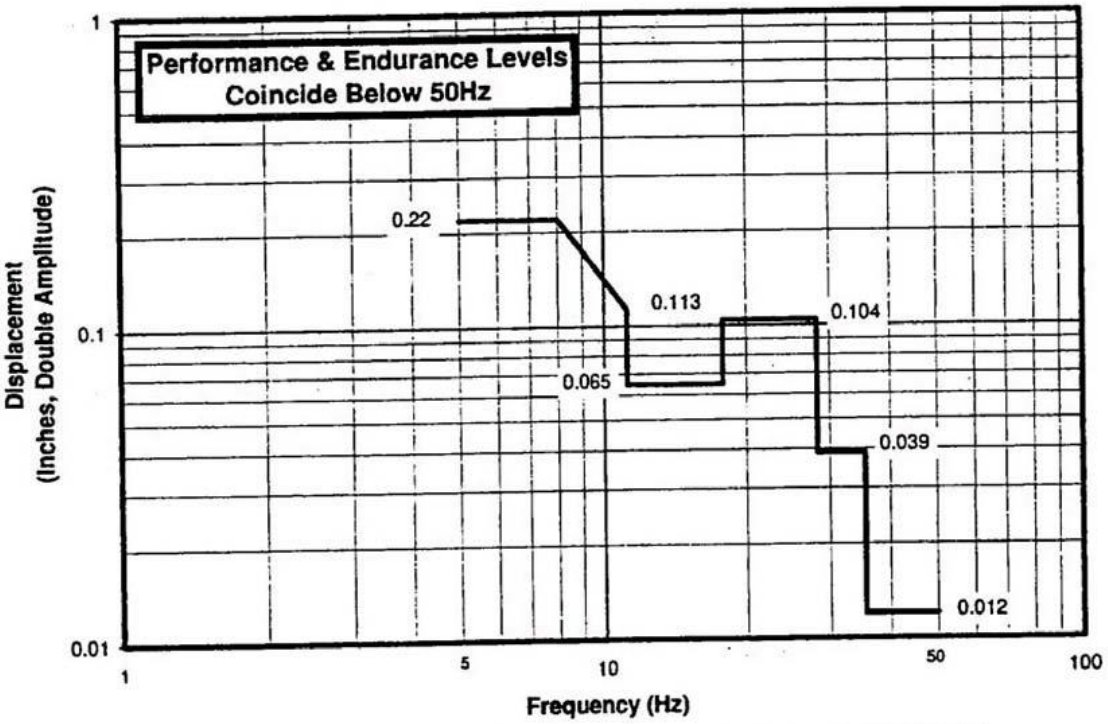
4.14.1.1 Sinusoidal vibration test. The ESCs shall be vibrated at 70 °F along the same orthogonal axis between 5 and 50 hertz and returning to 5 hertz in approximately 7.5 minutes for a total time of 20 minutes per axis, in accordance with figure 1. The test shall be repeated at -65 °F and again at +200 °F.

4.14.1.2 Random vibration test. The ESCs shall be vibrated along each orthogonal axis at 70 °F for 40 minutes per axis for a total of 120 minutes, in accordance with the applicable random vibration spectrum shape, endurance test levels of figure 2, and table VII. The test shall be repeated at -65 °F and then again at +200 °F.

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VIBRATION (NON-GUNFIRE) - RANDOM/SINUSOIDAL LOW FREQUENCY
SINUSOIDAL TEST LEVELS (5 - 50 Hz)
REGION V-FF-3

(Cockpit)



Non-Gunfire Vibration Region V-FF-3 Sinusoidal	
Frequency Range (Hz)	Displacement or Acceleration
5-8	0.22 in D.A.
8-11.2	±0.7 G
11.2-17.8	0.065 in D.A.
17.8-28.2	0.104 in D.A.
28.2-35.5	0.039 in D.A.
35.5-50	0.012 in D.A.

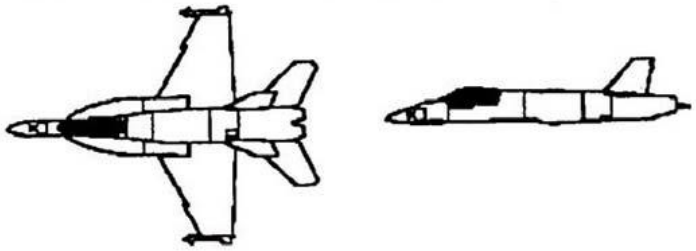
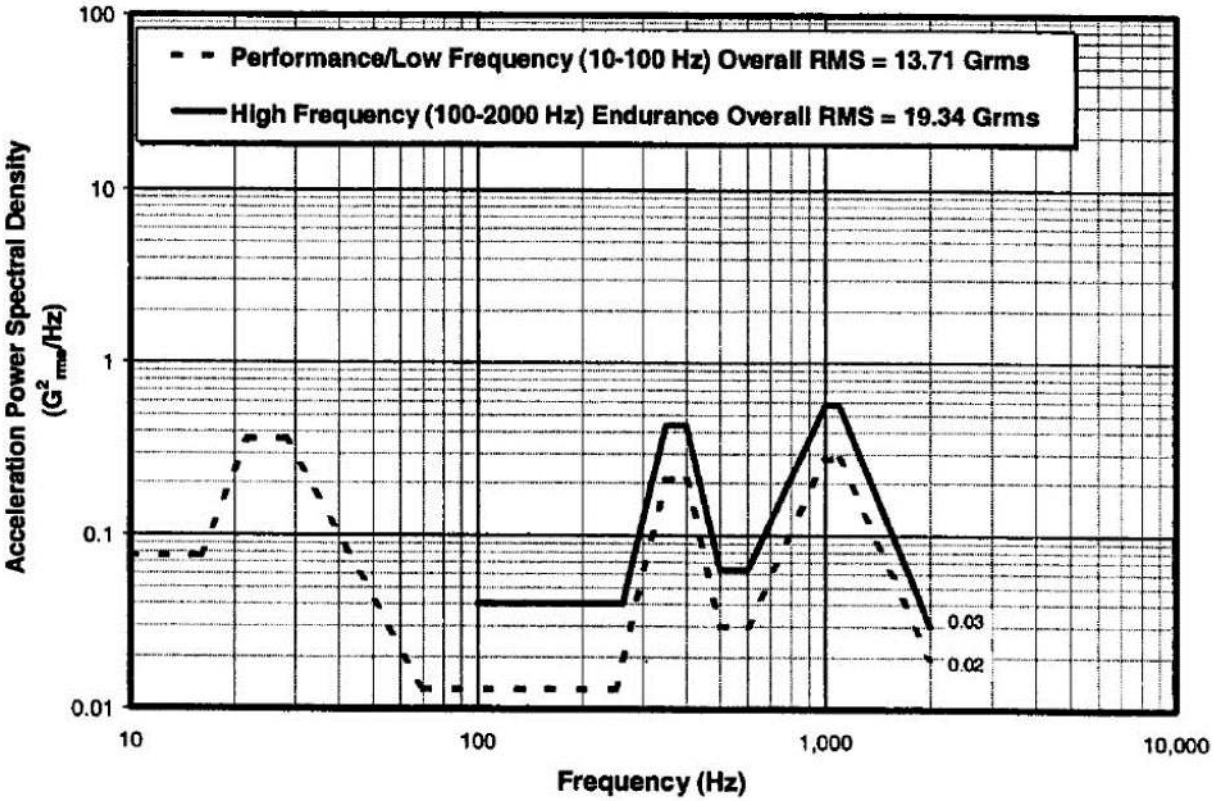


FIGURE 1. Low frequency vibration profile.

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Note: Use high frequency endurance levels.

FIGURE 2. Non-gunfire random vibration test profile.

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TABLE VII. Non-gunfire random vibration test levels.

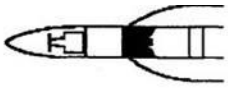
F/A-18 E/F Non-Gunfire Zone V-FF-3 Random			
Performance: Overall 13.71 G_{rms}		Endurance: Overall 19.34 G_{rms}	
Frequency Range (Hz)	$G2_{rms}$/Hz or dB/Octave	Frequency Range (Hz)	$G2_{rms}$/Hz or dB/Octave
10-16	0.076 G^2_{rms}/Hz	100-262	0.040 G^2_{rms}/Hz
16-22	13.992 dB/Octave	262-350	24.944 dB/Octave
22-28	0.363 G^2_{rms}/Hz	350-400	0.435 G^2_{rms}/Hz
28-70	-11.027 dB/Octave	400-500	-26.144 dB/Octave
70-250	0.013 G^2_{rms}/Hz	500-600	0.063 G^2_{rms}/Hz
250-350	24.944 dB/Octave	600-1,000	13.056 dB/Octave
350-400	0.211 G^2_{rms}/Hz	1,000-1,100	0.575 G^2_{rms}/Hz
400-500	-26.144 dB/Octave	1,100-2,000	-14.646 dB/Octave
500-600	0.030 G^2_{rms}/Hz	---	---
600-1,000	13.056 dB/Octave	---	---
1,000-1,100	0.279 G^2_{rms}/Hz	---	---
1,100-2,000	-14.646 dB/Octave	---	---
NOTE: Use endurance levels.			

4.14.2 Gunfire sinusoidal vibration. A sinusoidal vibration test at +70 °F is used in the frequency range of 50 to 2,000 hertz. Testing shall consist of logarithmic sweeps from 50 to 2,000 hertz and back to 50 hertz in 15 minutes plus fixed dwell periods in accordance with figures 3 and 4. The gunfire vibration tests shall be performed on each of the ESCs. Two accelerometers shall be used to monitor the response of the ESCs during the test. One accelerometer shall be located on a handle and the other on the body of one of the ESCs along the centerline of port number 4. The handle position for the testing shall be as specified in table III.

4.14.2.1 Vibration cycling. The items shall be vibrated along the same orthogonal axis, with the frequency varying at a logarithmic rate from 50 to 2,000 hertz and back to 50 hertz in 15 minutes. Double amplitude or vibratory acceleration levels shall be in accordance with figure 3.

4.14.2.2 Fixed dwell. The initiators shall be vibrated along the same orthogonal axis at the fixed narrow band dwell frequencies. The fixed narrow band dwells shall be performed by sweeping the frequencies about ±5 percent of the specified center frequency (see figure 4). The sweep rate shall be approximately the same as in 4.14.2.1. The vibration test levels shall be as shown on figure 4.

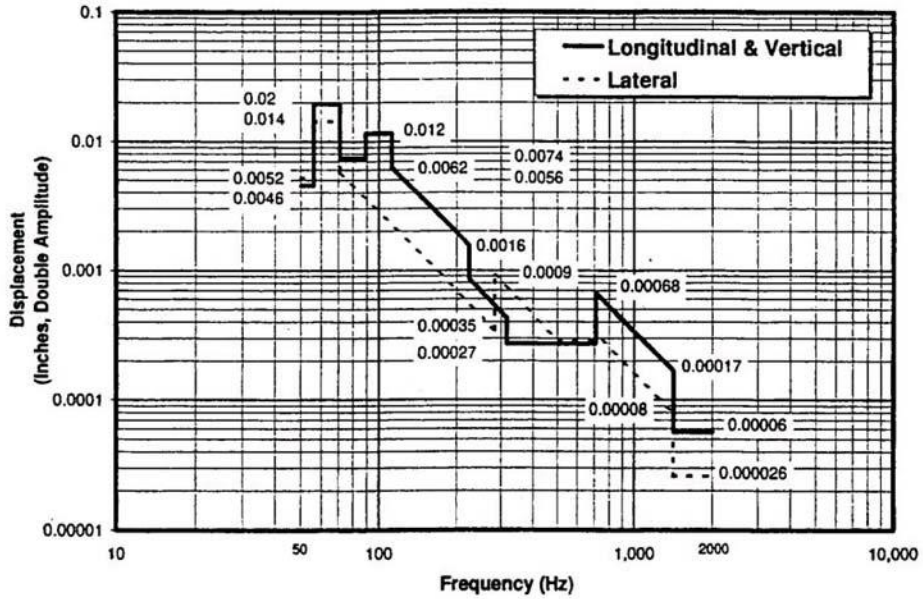
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VIBRATION (GUNFIRE) - SINUSOIDAL TEST LEVELS
REGION V-GF-5



(Forward Cockpit - Main Instrument Panel and Vertical Consoles)



Gunfire Vibration Region V-GF-5 Sinusoidal			
Longitudinal & Vertical		**Lateral**	
Frequency Range (Hz)	Displacement or Acceleration	Frequency Range (Hz)	Displacement or Acceleration
50-56 Hz	0.0046 in D.A.		
56-71 Hz	0.020 in D.A.	50-56 Hz	0.0052 in D.A.
71-89 Hz	0.0074 in D.A.	56-71 Hz	0.014 in D.A.
89-112 Hz	0.012 in D.A.	71-283 Hz	±1.4 G
112-224 Hz	±4.0 G	283-505 Hz	±3.7 G
224-315 Hz	±2.2 G	505-750 Hz	0.00029 in D.A.
315-707 Hz	0.00027 in D.A.	750-1414-Hz	±8.2 G
707-1414 Hz	±17.3 G	1414-2000 Hz	0.000026 in D.A.
1414-2000 Hz	0.000057 in D.A.		

FIGURE 3. Vibration (gunfire) – sinusoidal test schedule.

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Sweep time (min/axis)	60.0
Fixed Dwell (f_d) at the following Center frequencies (see note 1)	
67 Hz (63.7-70.4 Hz)	7.5
100 Hz (95-105 Hz)	7.5
135 Hz (128-142 Hz)	7.5
200 Hz (190-210 Hz)	15.0
267 Hz (254-280 Hz)	7.5
300 Hz (285-315 Hz)	7.5
335 Hz (318-352 Hz)	7.5
400 Hz (380-420 Hz)	15.0
467 Hz (444-490 Hz)	7.5
500 Hz (475-525 Hz)	7.5
Total Time Each Axis (minutes)	150.0

FIGURE 4. Vibration (gunfire) – sinusoidal test schedule with time.

4.15 Shock. First article test specimens specified in table III shall be exposed to a service shock environment in accordance with MIL-D-21625. The handle position for the testing shall be as specified in table III. The shock pulse waveform shall be a terminal peak saw tooth. The peak amplitude shall be 20 g's and the duration shall be 11 milliseconds. Three shocks shall be applied in each direction of the three mutually perpendicular axes, for a total of 18 shocks.

4.16 Ballistic test. The ESCs shall be selected from each production lot in accordance with 4.4.1. The sample of ESCs shall be divided, as nearly as possible, into three equal groups. If the sample of ESCs is not equally divisible by three, the excess of ESCs shall be placed in the group to be fired with the handle in the normal position. One group shall be fired with the handle in the normal position, one group shall be fired with the handle in the solo position, and one group shall be fired with the handle in the aft position. The group to be fired with the handle in the solo position shall have a solo spacer in accordance with NAVAIR Drawing 30003-850AS853 installed. Each group shall be subdivided, as nearly as possible, into three equal subgroups. If the sample of ESCs is not equally divisible by three and there is one excess ESC, the excess ESC shall be placed in the subgroup to be fired at -65 °F. If the sample of ESCs is not equally divisible by three and there are two excess ESCs, the excess ESCs shall be split between the hot and cold subgroups, with one ESC placed in the subgroup to be fired at -65 °F and one ESC placed in the subgroup to be fired at +200 °F. One subgroup shall be fired at -65 °F, one subgroup shall be fired at +70 °F, and one subgroup shall be fired at +200 °F. One subgroup each shall be conditioned, with the test tips installed, for a period of time neither less than 4 hours nor greater than 24 hours at -65±5 °F and +200± 5 °F. The remaining subgroup shall be conditioned, with the test tips installed, for a period of time not less than 4 hours at +70±5 °F. After temperature conditioning, the handle shall move freely between the normal, solo, and aft positions. The ESC shall be fired in the test fixture of NAVAIR Drawing 30003-850AS804 (as shown on NAVAIR Drawing 30003-850AS135) within 5 minutes after removal from the temperature conditioning chamber. If any ESC is exposed to the temperature environment for greater than the time allowed or not fired within 5 minutes after removal, it shall be returned to a +70±5 °F environment for at least 4 hours and then re-exposed to the original environment for not less than 4 hours and no more than 24 hours before test firing. The re-exposure may be performed as many times as necessary to complete the required ballistic test. Instrumentation shall be provided to measure and record the high-order detonation of the designated ESC output SMDC test tip. The production lot shall be accepted if the sample fulfills the criteria of 4.16.1.

4.16.1 Acceptance criteria.

4.16.1.1 High-order detonation. The high-order detonation of the output SMDC test tip propagated, as specified in 3.7.1, shall be demonstrated in a 6061-T6 aluminum witness plate. The lot shall be acceptable with respect to high-order detonation if the ballistic sample meets the requirements of 3.7.1. Any defects shall be cause for rejection.

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4.16.1.2 Misfire. Any ESC that does not fire under the conditions specified in 4.16 shall be considered defective and the lot considered unacceptable, unless the misfire is clearly attributable to faulty test equipment or procedure.

4.16.2 Retest. There shall be no retests.

4.16.3 Test failure. If test failure is attributable to an assignable cause, excluding the test ESC, the original test results shall be discarded and that test reconducted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The ESC, DCU-241/A was designed to select the ejection mode (normal, solo, or aft) for the F-18, T-45, and AV-8 aircraft aircrew escape systems. Since the ESC was designed strictly for use in military aircraft, there is no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification and source control NAVAIR Drawing 850AS135.

b. When first article inspection is required (see 3.1 and 4.2).

c. Materials required as established by the technical service release (see 3.3).

d. Items of data required for each first article and production lot.

e. Direction for shipping radiographic plates of entire production lot along with ballistic sample (see 3.7) to the activity conducting production lot acceptance inspection (see 4.3).

f. Production lot size and test activity (see 4.4.1).

g. Direction for shipping radiographic plates of entire production lot along with ballistic sample to the activity conducting production lot acceptance test (see 4.8.1 and 4.8.2).

h. Packaging requirements (see 5.1 and 6.3).

i. Safety precautions (see 6.4).

6.3 Packaging. Packaging should be in accordance with SPI-PHST-249.

6.4 Safety precautions. The safety precautions requirements of DoDM 4145.26, DOD Contractor's Safety Manual for Ammunition and Explosives, are applicable and should be specified in the contract as required by the Defense Acquisitions Regulation (DAR) 1-323. When this specification is used as part of the description of work to be accomplished by a government activity, the safety precaution requirements of NAVSEA OP-5, Volume 1, Ammunition and Explosives Ashore, should be made applicable.

6.5 Explanation of terms. For the purpose of this document, NAVSUPINST 4030.28, Packaging of Materiel, when applicable, should be used to clarify terms.

6.6 Subject term (key word) listing.

Aircrew escape system
AV-8
F-18
T-45

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6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 1377-2023-003)

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